

Photosynthesis Lab: Energy Flow in a Leaf

California CCSS Mathematics Standards Alignment

OBJECTIVES	CODE ASSIGNMENTS	HIGH CONFIDENCE	GRADE BAND
8 6 aligned · 2 not	11 11 distinct standards	1 6 med · 4 low	— inferred from context

Auto-curated via run_alignment.py (heuristic mode, no LLM in the loop).

Section 1: Setup and Background

Watch Me

Watch the introductory video on photosynthesis and cellular respiration.

sections[0].objectives[0]

Note: logistical/operational objective, no MATH standard applies

logistical/operational objective, no MATH standard applies

Equation Review

Write out the balanced chemical equation for photosynthesis: $6\text{CO}_2 + 6\text{H}_2\text{O} + \text{light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. Identify reactants and products.

sections[0].objectives[1]

A-REI.C.5

A

categorized-standardsCaliforniaCA-MATH

Grade 9-12

Solve systems of equations.

70 / 100

MEDIUM

Prove that, given a system of two equations in two variables, replacing one equation by the sum of that equation and a multiple of the other produces a system with the same solutions.

WHY THIS MAPS

keyword boost applied; matched rule: \b(equation|inequal|solv(e|ing))linear equation|sy

A-REI.B.4.a

A

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Grade 9-12

Solve equations and inequalities in one variable.

70 / 100

MEDIUM

Use the method of completing the square to transform any quadratic equation in x into an equation of the form $(x - p)^2 = q$ that has the same solutions. Derive the quadratic formula from this form.

WHY THIS MAPS

keyword boost applied; matched rule: \b(equation|inequal|solv(e|ing))linear equation|sy

7.EE.A.2

EE

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Grade 7

Use properties of operations to generate equivalent expressions.

70 / 100

MEDIUM

Understand that rewriting an expression in different forms in a problem context can shed light on the problem and how the quantities in it are related. For example, $a + 0.05a = 1.05a$ means that “increase by 5%” is the same as “multiply by 1.05.”

WHY THIS MAPS

keyword boost applied; matched rule: \b(equation|inequal|solv(e|ing))linear equation|sy

Section 2: Experimental Design

Hypothesis

Develop a testable hypothesis predicting how oxygen production rate changes with light intensity. Justify your prediction using the photosynthesis reaction.

sections[1].objectives[0]

F-LE.A.1.b

F

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Grade 9-12

Construct and compare linear, quadratic, and exponential models and solve problems.

50 / 100

LOW

Recognize situations in which one quantity changes at a constant rate per unit interval relative to another. ■

WHY THIS MAPS

matched rule: \b(reason(ing)?|argument|critique|justif(y)|cation

Variables

Identify the independent variable, dependent variable, and at least three controlled variables for the experiment. Explain why each control matters.

sections[1].objectives[1]

6.EE.C.9

EE

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Grade 6

Represent and analyze quantitative relationships between dependent and independent variables.

90 / 100
HIGH

Use variables to represent two quantities in a real-world problem that change in relationship to one another; write an equation to express one quantity, thought of as the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation. For example, in a problem involving motion at constant speed, list and graph ordered pairs of distances and times, and write the equation $d = 65t$ to represent the relationship between distance and time.

WHY THIS MAPS

keyword boost applied; matched rule: $\backslash b(expression|variable|coefficient|term|simplify|d$

Procedure and Data Collection

Plan and conduct an investigation that measures oxygen bubble production from Elodea sprigs at five different light intensities. Record data in a structured table with units.

sections[1].objectives[2]

6.SP.B.5.b

SP

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Grade 6

Summarize and describe distributions.

70 / 100
MEDIUM

Describing the nature of the attribute under investigation, including how it was measured and its units of measurement.

WHY THIS MAPS

keyword boost applied; matched rule: $\backslash b(data|graph|chart|plot|histogram|scatter[-]?plo$

6.SP.B.5.d

SP

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Grade 6

Summarize and describe distributions.

70 / 100
MEDIUM

Relating the choice of measures of center and variability to the shape of the data distribution and the context in which the data were gathered.

WHY THIS MAPS

keyword boost applied; matched rule: $\backslash b(data|graph|chart|plot|histogram|scatter[-]?plo$

S-ID.A.4

S

categorized-standards|California|CA-MATH

Grade 9-12

Summarize, represent, and interpret data on a single count or measurement variable.

70 / 100
MEDIUM

Use the mean and standard deviation of a data set to fit it to a normal distribution and to estimate population percentages. Recognize that there are data sets for which such a procedure is not appropriate. Use calculators, spreadsheets, and tables to estimate areas under the normal curve. ■

WHY THIS MAPS

keyword boost applied; matched rule: $\backslash b(data|graph|chart|plot|histogram|scatter[-]?plo$

Producer Role

Explain how plants act as producers in a food web. Trace the path of carbon and energy from sunlight through a producer, to a primary consumer, to a secondary consumer.

sections[2].objectives[0]

N-VM.C.10

N

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Grade 9-12

Perform operations on matrices and use matrices in applications.

50 / 100
LOW

Understand that the zero and identity matrices play a role in matrix addition and multiplication similar to the role of 0 and 1 in the real numbers. The determinant of a square matrix is nonzero if and only if the matrix has a multiplicative inverse.

WHY THIS MAPS

TF-IDF top match

Carbon Cycle Diagram

Construct a diagram of the carbon cycle showing how photosynthesis, cellular respiration, decomposition, and combustion move carbon between the atmosphere, biosphere, hydrosphere, and geosphere.

sections[2].objectives[1]

2.MD.B.6

MD

categorized-standards\California\CA-MATH

Grade 2

Relate addition and subtraction to length.

50 / 100
LOW

Represent whole numbers as lengths from 0 on a number line diagram with equally spaced points corresponding to the numbers 0, 1, 2, . . . , and represent whole-number sums and differences within 100 on a number line diagram.

WHY THIS MAPS

TF-IDF top match

3.NF.A.2

NF

categorized-standards\California\CA-MATH

Grade 3

Develop understanding of fractions as numbers.

50 / 100
LOW

Understand a fraction as a number on the number line; represent fractions on a number line diagram.

WHY THIS MAPS

TF-IDF top match

Section 4: Reflection

Lab Reflection

Write a 100+ word reflection on what surprised you about the lab results, what sources of error you noticed, and how you would refine the experiment if you ran it again.

sections[3].objectives[0]

Note: reflective/metacognitive objective, no MATH standard applies

reflective/metacognitive objective, no MATH standard applies